Silicon hybrid photonic crystal nanocavity formed by printing a single InGaAsP nanowire

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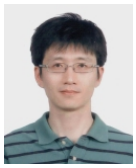
Ju-Young Kim*, Seungwoo Lee, Jae-Hyeon Son, Yong-Hee Lee, Bumki Min

1: Dep. of Mechanical Engineering, KAIST, Daejeon, South Korea

2: Dep. of Physics, KAIST, Daejeon, South Korea

*peterfish.jy@gmail.com

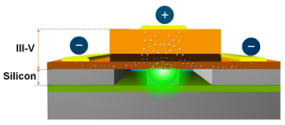




Abstract

Abstract A single InGaAsP nanowire generates an optical cavity on silicon photonic crystal waveguide as well as providing an optical gain. We demonstrated a transfer printing of a single InGaAsP nanowire onto the silicon photonic crystal waveguide at a desired place. The hybrid structure could provide a greater flexibility in the strategies for an efficient silicon hybrid laser.

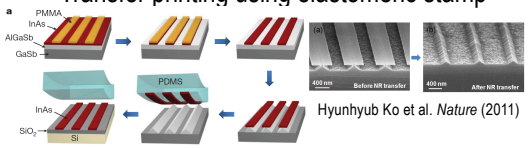
Laser on Silicon: Heterogeneous integration



Hybrid 3-5-on-Si lasers are currently considered as the most practical lasers on silicon.

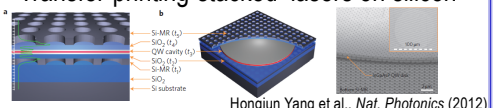
Di Liang et al. *Nat. Photonics* (2011)

Transfer printing using elastomeric stamp



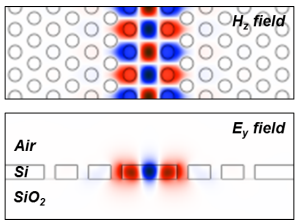
Hyunhyub Ko et al. *Nature* (2011)

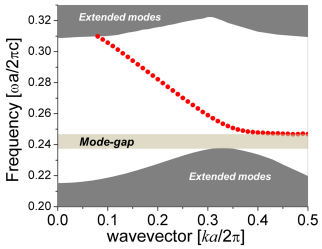
Transfer-printing-stacked lasers on silicon



Hongjun Yang et al., *Nat. Photonics* (2012)

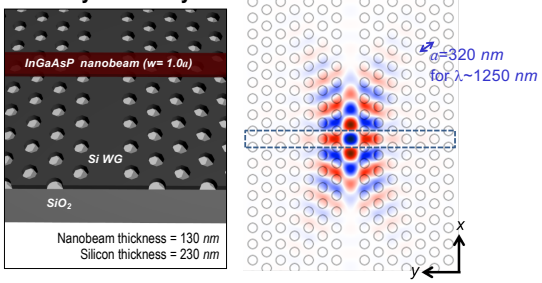
Silicon hybrid photonic crystal nanocavity





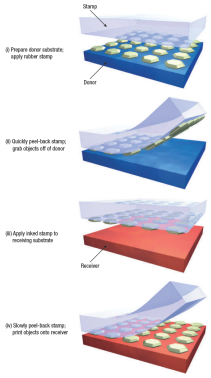
We investigate the silicon PhC WG by using 3-D FDTD computation. The dispersion has a cutoff frequency at 0.2468 and no photon can propagate along the PhC WG below the cutoff. The external perturbation, which causes the frequency of the WG mode below the cutoff, allows a cavity mode to be excited at the specific site.

Optical cavity formed by a Nanowire on Silicon PhC WG

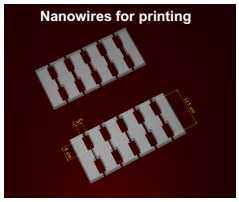


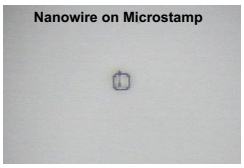
FDTD computation predicts that a resonant mode with a quality-factor of 450 is achieved due to the incorporation of the InGaAsP nanobeam into silicon PhC WG.

Transfer printing of Single Nanowire

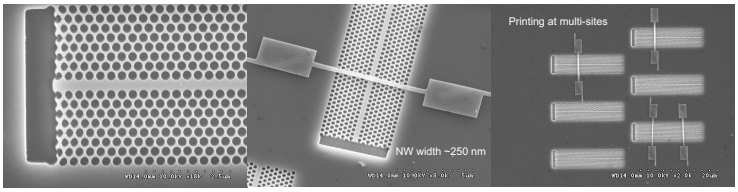


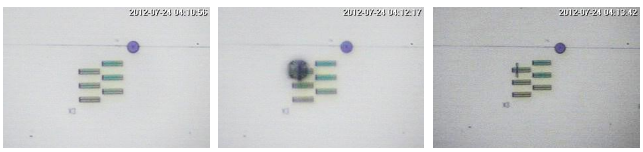
M.A.Meitl et al., *Nat. Mater.* (2006)





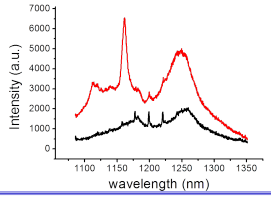
Printed InGaAsP Nanowires on silicon photonic crystal waveguides





The single nanowire can be printed at desired location on silicon photonic crystal waveguide by using an elastomeric microstamp. SEM and optical microscopic images.

Preliminary PL experiment



- Photoluminescence (PL) of the single nanowire was performed at room temperature with Ti:Sapp fs-laser pumping.
- Resonance peaks were observed by the transfer printing on the Si photonic crystal waveguide.
- Low temperature PL experiment is needed due to the excessive heat on the nanowire.

Summary

We propose the silicon photonic crystal nanocavity which is formed by placing an InGaAsP nanobeam. The nanobeam makes a silicon WG mode to be shifted below the cutoff and forms a resonant cavity on silicon chip. By taking advantage of the reliable techniques for the integration of InGaAsP nanobeam, one could simultaneously generate both an optical nanocavity and optical gain onto a desired location of the silicon PhC WG at high throughputs. This result could be a new route to realize an efficient hybrid nanolaser for the silicon photonic integrated circuit (PIC).

KAIST KOREA ADVANCED INSTITUTE OF SCIENCE AND TECHNOLOGY 한국과학기술원 SINCE 1971

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